

iDEC Responsible Research Form

Dear iDEC Teams,

This is the iDEC 2026 Responsible Research Form. Submission of this form is mandatory for your participation in iDEC. Please remember: safety and ethical considerations are an important part of modern research and we are not asking you these questions to waste your time, but to educate you, and protect you and those around you! Please take the time to answer the questions as a team, and if you need help, do not hesitate to contact the Commission for Responsible Research (vinke@idechq.org). We are here to help you with things like this!

The Responsible Research Form should be submitted prior to the start of the wetlab, but no later than July 1. If you find that you need to change parts of your projects (e.g., use additional genes), please contact us and we will approve those changes as soon as possible.

The responsible research form consists of 5 sections: General Information about your team, Laboratory Biosafety, Genes and Chassis, Biosecurity, and Ethical Considerations. We have gathered resources for each section to guide your team, and we have completed the entire form as an example on our website

(https://idec.io/pages/ethics_biosafety_and_biosecurity.html).

Be aware that you are able to shape the values of your research community. In our ideal directed evolution research community, safety, security, and ethical considerations are valued and common place. We hope the iDEC responsible research project can teach you how important these values are. We hope you learn more about your project as you think about biosafety, biosecurity, and ethical considerations, and have a wonderful time during iDEC 2026!

Your Commission for Responsible Research

***Required**

1. Team Name *

VCA-Prudens

2. Email address for questions regarding your Responsible Research Form *

mo202505@163.com

3. What is the research hypothesis of your project? *

In the field of synthetic biology, the manipulation of large DNA fragments has long been a critical challenge. In recent years, the discovery of ISCro4 has provided a promising new tool to address this problem. We aim to improve the efficiency of ISCro4 through directed evolution, thereby facilitating the manipulation of large DNA fragments.

4. Please provide a detailed abstract of your project. *

Manipulation of DNA sequences is the core technology of biotechnology, especially synthetic biology. While CRISPR/Cas systems made small-scale genome editing very efficient, large DNA sequences manipulation poses a major challenge. Recently discovered ISCro4 is an advanced tool for large-scale genomic engineering but it still requires further optimization of its efficiency. This project proposes the use of directed evolution approach to improve ISCro4 editing efficiency. First of all, we are going to develop the method of quantifying editing efficiency by coupling DNA fragment deletion with antibiotic resistance restoration in *Escherichia coli*. Using structural information about ISCro4 protein, we are going to make saturation mutagenesis and combine mutations found in hotspots to improve ISCro4 activity. Finally, the best variants of ISCro4 obtained from directed evolution are going to be tested on the ability to induce large deletions, insertions and inversions of DNA sequences. Thus, our goal is to obtain highly-efficient and stable variant of ISCro4 which can reliably manipulate large DNA fragments. This work will expand the genome engineering toolbox and provide a powerful platform for synthetic biology and other applications requiring precise large-scale DNA editing.

State-of-knowledge

The following four questions will ask you how much you knew about risk assessment before iDEC. You don't have to answer them, but they would help our committee design better, more tailored educational projects. There are no wrong answers to these questions and it is okay if you answer "no" to all of them - it is not your fault if no one told you about these risks.

5. Have you ever assessed biosafety risks of a scientific project? When was the first time and place someone taught you about biosafety risks?

The first time I learned systematic biosafety risk knowledge came from standardized biosafety training courses, safety education lecture hall before students were allowed to enter wet labs. The core contents covered: classification of biological hazards, risk grading of microorganisms, personal protective equipment operation rules, waste disposal, emergency response for leakage or contamination, and ethical & biosafety boundaries for genetic engineering modified organisms. This formal training laid the foundation for me to judge biosafety risks of various scientific projects later.

6. Have you ever assessed biosecurity risks of a scientific project? When was the first time and place someone taught you about the difference between biosafety and biosecurity risks?

Absolutely. I have assessed biosecurity risks for synthetic biology projects by identifying dual-use research hazards and risks of malicious misuse of biological technologies.

I first learned the difference between biosafety and biosecurity at an undergraduate seminar in my university's biomedical teaching building. The trainer explained that biosafety deals with accidental lab accidents, while biosecurity covers intentional misuse of biological resources and technologies, which helped me classify risks properly in all later project assessments.

7. Do you know what "Dual-Use Research of Concern " means? Where and when did you learn about dual-use research of concern?

Absolutely, I know DURC well. It describes beneficial life science research whose knowledge, strains or technologies could be intentionally misused to trigger major public health or security threats, such as engineering more dangerous pathogens or weaponizing toxins.

I first learned this concept in a university biosecurity workshop in my sophomore year, where instructors explained its definition, real-world cases and risk assessment standards. Afterwards, I further practiced DURC screening while finishing the biosecurity evaluation.

8. Do you know what "Gain of function experiment" means? Where and when did you learn about them?

Yes, I clearly understand gain-of-function (GoF) experiments. They are genetic modifications that give organisms new or strengthened biological features. Most GoF work like engineering microbial production strains is safe, but pathogen-focused GoF that boosts transmissibility or virulence is classified as DURC with severe biosafety and biosecurity risks.

I first learned this concept in a biosecurity seminar at my university's life science building during my second undergraduate year. The trainer linked risky GoF experiments to dual-use threats, and I further practiced identifying GoF hazards when finishing the safety evaluation of our project.

Laboratory Biosafety

Thank you for helping us by sharing your level of knowledge prior to iDEC! We will now start with the risks you have identified for your iDEC project. The first category of risks is "Laboratory Biosafety." In this section, you will need to indicate that your lab meets the criteria to be a safe place for you to work, and ensure no accidental release of your GMOs.

Biosafety describes personal protection and protection from accidental release of biological agents that have the potential to harm plants, animals, humans, or the environment. In this laboratory biosafety section of the Responsible Research Form, your teams will need to answer questions about safety equipment in your lab and safety countermeasures to prevent accidental release of GMOs.

If you are looking for resources or would like to learn more, we recommend you take a look at the WHO Laboratory Biosafety Manual:

<https://www.who.int/publications/i/item/9789240011311>

9. Mention the PI of your team and the relevant qualifications of your PI to function as your team's person responsible for safety. *

Our group's Principal Investigator is Dr. Jian Mo. He earned his PhD in Biochemistry & Molecular Biology and has 10+ years of experience in lab research and management. Additionally, he has undergone training in bio-safety and has achieved certification in laboratory safety management. In his capacity as the safety officer of the group, Dr. Mo will oversee the development of safety protocols, train group members on biosafety measures, conduct risk assessments, and ensure that all our lab practices comply with appropriate standards and guidelines.

10. Did you get permission from your institute to perform the experiments necessary for your iDEC project? *

Yes, I have got permission from my institute.

11. List all topics mentioned in your safety introduction. *

Biosecurity risk assessment of scientific research projects
 Core differences between biosafety and biosecurity
 Biosafety: Accidental lab hazards, unintentional contamination, environmental release of modified organisms
 Biosecurity: Intentional malicious misuse, theft of biohazard materials, dual-use threats
 Dual-Use Research of Concern (DURC)
 Definition of DURC
 Typical high-risk DURC research activities
 Linkage between DURC and biosecurity governance
 Gain-of-Function (GoF) experiments

Definition of gain-of-function modification
 Two categories of GoF experiments (low-risk industrial GoF / high-risk pathogen GoF)
 Biosafety risks of high-risk GoF
 Biosecurity risks of high-risk GoF (classified as DURC)
 Hazard classification of microorganisms & biological hazards
 Laboratory containment level standards
 Personal protective equipment (PPE) operation specifications
 Biological waste disposal rules
 Emergency response for liquid leakage, biological contamination accidents
 Genetic engineering ethics and safety boundaries
 Institutional Biosafety Committee (IBC) guidance & training

12. What is the biosafety level of your laboratory? *

Mark only one oval.

- ☐ We have a computational project and don't use a laboratory
☒ BSL1
☐ BSL2
☐ BSL3
☐ Other: _____

13. Describe the safety measures in your laboratory (e.g. eye shower, emergency shower, etc) *

Our lab has complete emergency and biosafety equipment: eye wash station and full-body emergency shower for splash accidents; Class II biosafety cabinets for microbial operations; autoclaves to sterilize bio-waste. We also have fire extinguishers, spill kits, fume hoods, color-coded biohazard bins, PPE including goggles and lab coats, and hand-washing sinks. All benches are equipped with disinfectants for routine surface sterilization.

14. Describe how GMOs are prevented from being unintentionally released into the environment, including waste management *

Multiple layered measures block unintended GMO environmental release, centered on lab containment and full sterilization waste management. Inside the lab, negative-pressure biosafety cabinets, sealed culture vessels and routine surface disinfection prevent aerosol and liquid spills of GMOs. Staff wear full PPE and follow strict access rules to avoid carrying modified organisms out accidentally. For waste: All solid GMO-contaminated waste is sealed in biohazard bags and fully autoclaved to kill viable GMOs. Liquid cultures are soaked in strong disinfectant for 24 hours before drainage. All residual GMO materials are completely inactivated before leaving the lab. Additionally, GMO strains are stored in locked sealed containers, transport off-lab is restricted, and no open environmental cultivation of engineered microbes is performed to cut off all escape pathways.

15. Do you work with animals or samples (urine, blood, saliva, etc) derived from animals or humans? *

No

16. Are you working with chemicals that are mutagenic, carcinogenic, explosives, or narcotics? If yes indicate what chemicals, in what amounts you use them, and what safety measures are in place. *

We use several commonly used antibiotics for routine molecular cloning experiments. Although the use of antibiotics may contribute to the selection of antibiotic-resistant mutants, all antibiotic-containing media and waste will be handled and disposed of following strict laboratory biosafety procedures to minimize this risk. In addition, the antibiotics used in our project are standard laboratory antibiotics that are widely used in molecular biology research. Therefore, we do not anticipate that their use in this project will pose significant biosafety concerns.

Genes and Chassis

Genes and chassis of each team must be biosafety level 1 or 2. Any genes and chassis from organisms that fall into the biosafety level 3 or 4 category are prohibited to be used in iDEC. There will be a whitelist of chassis and all genes and chassis used beyond that must be listed here.

Whitelists are kind of outdated when it comes to synthetic biology, since the taxonomic classification of organisms into safety groups does not necessarily reflect the risks of an organism to cause harm (e.g., an organism might carry genes for toxins that would not be expressed under the conditions used to characterize that organism, so the recommended biosafety level would be low, but we can still derive the toxin from that organism and express it recombinantly). That said, we decided to provide a short whitelist for iDEC so you don't have to list the usual lab workhorses like your *Escherichia coli* lab strain. Our whitelist includes the FDA's GRAS organism as well as commonly used organisms for directed evolution projects. The iDEC whitelist can be found in this form and on the iDEC website and any organisms, genes from this white list do not need to be listed here unless you are evolving them (https://idec.io/pages/ethics_biosafety_and_biosecurity.html).

For us, it is very important that you really think about what additional risks might come from your evolved molecule (or pathway, or genome). The thing about directed evolution is that you are creating something novel, and that novel product may not have the same level of safety as the scaffold you started with.

17. List all chassis you plan to use that are not listed in the iDEC whitelist below, as well as their biosafety level (with reference) and a short explanation how you plan to use these. *

<i>Acetobacter suboxydans</i>	<i>Eisenia bicyclis</i>	<i>Lactococcus lactis</i>	<i>Porphyra suborbiculata</i>
<i>Acetobacter xylinum</i>	<i>Endothia parasitica</i>	<i>Laminaria angustata</i>	<i>Porphyra crispata</i>
<i>Actinoplane missouriensis</i>	<i>Eremothecium ashbyii</i>	<i>Laminaria claustronia</i>	<i>Porphyra tenera</i>
<i>Anallipus japonicus</i>	<i>Escherichia coli</i> BL21 (DE3)	<i>Laminaria digitata</i>	<i>Rhizopus niveus</i>
<i>Aspergillus niger</i>	<i>Escherichia coli</i> DH5alpha	<i>Laminaria japonica</i>	<i>Rhizopus oryzae</i>
<i>Aspergillus niger</i> var. Awamori	<i>Escherichia coli</i> DH5alpha F'	<i>Laminaria longiruris</i>	<i>Rhodomenis palmata</i>
<i>Aspergillus oryzae</i>	<i>Escherichia coli</i> TOP10	<i>Laminaria longissima</i>	<i>Saccharomyces cerevisiae</i>
<i>Arabidopsis thaliana</i> WT	<i>Eucheuma cottonii</i>	<i>Laminaria ochotensis</i>	<i>Saccharomyces fragilis</i>
Cooney et Emerson	<i>Eucheuma spinosum</i>	<i>Laminaria saccharina</i>	<i>Scytosiphon lome</i>
<i>Bacillus cereus</i>	<i>Furcellaria fastigiata</i> of the class	<i>Leuconostoc citovarum</i>	<i>Streptococcus cremoris</i>
<i>Bacillus coagulans</i>	<i>Rhodophyceae</i> (red seaweed)	<i>Leuconostoc dextranicum</i>	<i>Streptococcus lactis</i> subspecies
<i>Bacillus licheniformis</i>	<i>Fusarium moniliforme</i>	<i>Leuconostoc mesenteroides</i> strain	<i>diacetylactis</i>
<i>Bacillus stearothermophilus</i>	<i>Gigartina acicularis</i>	NRRL B-512(F)	<i>Streptococcus lactis</i>
<i>Bacillus subtilis</i>	<i>Gigartina pistillata</i>	<i>Macrocystis pyrifera</i>	<i>Streptococcus thermophilus</i>
brown algae	<i>Gigartina radula</i>	<i>Mortierella vinaceae</i> var.	<i>Streptomyces chattanoogensis</i>
<i>Candida guilliermondii</i>	<i>Gigartina stellata</i>	<i>Raffinoseutilizer</i>	<i>Streptomyces griseus</i>
<i>Candida lipolytica</i>	<i>Gloiopeltis furcata</i>	<i>Mucor miehei</i>	<i>Streptomyces natalensis</i>
<i>Candida pseudotropicalis</i>	<i>Hizikia fusiforme</i>	<i>Mucor miehei</i> var. Cooney et Emerson	<i>Streptomyces olivaceus</i>
<i>Candida utilis</i>	<i>Kjellmaniella gyrate</i>	<i>Mucor pusillus</i> Lindt	<i>Streptomyces olivochromogenes</i>
<i>Chlamydomonas reinhardtii</i>	<i>Kluyveromyces lactis</i>	<i>Penicillium roquefortii</i>	<i>Streptomyces rubiginosus</i>
<i>Chondrus crispus</i>	<i>Kluyveromyces marxianus</i> var. <i>lactis</i>	<i>Petalonia fascia</i>	<i>Xanthomonas campestris</i>
<i>Chondrus ocellatus</i>	<i>Lactobacillus bulgaricus</i>	<i>Porphyra deusta</i>	
	<i>Lactobacillus fermentum</i>	<i>Porphyra perforata</i>	

Only use *Escherichia coli* DH5alpha chassis strains is classified as Risk Group 1 (BSL-1) with zero pathogenic potential. All experimental work strictly follows GMO containment rules to prevent unintended environmental release; no field trials or open-air cultivation are performed. No BSL-2/3/4 organisms are involved in the entire project, complying with iDEC competition safety regulations.

18. List all genes and plasmids you plan to use that are not derived from one of the organism on the iDEC whitelist above, as well as their original host/lab and their biosafety level of their original host (with references) and a short explanation on how you plan to use these. *

ISCro4. It is from *Citrobacter rodentium*, a Gram-negative bacterium that naturally infects mice to model human gastrointestinal diseases. Although often misidentified as a BSL-1 agent due to its host specificity, it is officially classified as a **Biosafety Level 2 (BSL-2)** pathogen because it can cause disease in mammals. (<https://www.atcc.org/products/baa-352>)

ISCro4 is a protein encoded within a transposon that catalyzes the movement of large DNA fragments, including their excision, integration, and inversion. In our project, ISCro4 serves as the target of directed evolution. We aim to enhance its efficiency in mediating DNA excision, integration, and inversion, thereby improving large-fragment genome engineering and advancing the development of biotechnology.

19. Which genes are evolved in your experiment? What kind of improvement do you want to achieve by your evolution (e.g. increase activity, increase stability)? *

ISCro4 is a protein encoded within a transposon that catalyzes the movement of large DNA fragments, including their excision, integration, and inversion. In our project, ISCro4 serves as the target of directed evolution. We aim to enhance its efficiency in mediating DNA excision, integration, and inversion, thereby improving large-fragment genome engineering and advancing the development of biotechnology. It is from *Citrobacter rodentium*, a Gram-negative bacterium that naturally infects mice to model human gastrointestinal diseases. Although often misidentified as a BSL-1 agent due to its host specificity, it is officially classified as a **Biosafety Level 2 (BSL-2)** pathogen because it can cause disease in mammals. (<https://www.atcc.org/products/baa-352>). This enzyme is only used to manipulate DNA and there are no other hidden dangers.

Biosecurity

Biosecurity describes the prevention of biological agents from unauthorized access, theft, misuse or loss. In the biosecurity section of the responsible research form, your team is required to evaluate potential biosecurity risks and their countermeasures. In this section we will ask for the dual-use research of concern potential. Dual-use research of concern in the life sciences, describes the potential of your project, generated technologies and knowledge to be directly misused by someone else to cause serious harm to animals, humans, plants, national security or the environment. Please take this section seriously. We have seen a lot of teams becoming very creative when it comes to envision 10,000 benefits of their project, but we almost never become creative when it comes to envision the risks. That is something we need to change! For most risks you identify, you will notice that countermeasures can be easily integrated and that makes your project better and more likely to be used in a project beyond your labbench!

If you would like to learn more, take a look at the WHO biorisk management guidance (https://www.who.int/ihr/publications/WHO_CDS_EPR_2006_6/en/), the NIH dual-use research of concern information page (<https://osp.od.nih.gov/biotechnology/dual-use-research-of-concern/>) or attend the free FutureLearn "Next Generation Biosecurity: Responding to 21st Century Biorisks" course (<https://www.futurelearn.com/courses/biosecurity>).

20. Describe the access control measures of your laboratory. *

Our laboratory adopts facial recognition system as the core access control measure. Only registered staff with valid facial information can unlock the entrance and gain access. Unregistered visitors are prohibited from entering freely. They need to make advance applications and be accompanied by internal laboratory personnel before entry. All entry and exit records are automatically logged by the system to track personnel movement in real time. Traditional locks and keys are still used in our laboratory. These are usually high - security locks that require a specific key to open. The keys are distributed only to a limited number of authorized personnel.

21. Evaluate if your project has Dual Use Research of Concern potential. *

Our project does not have this concern.

22. If yes, indicate what measures your team takes to minimize the risk of misuse. *

N.A.

Ethical Considerations

Identifying ethical issues of scientific research projects is a big part of responsible research. To be able to identify possible unethical practices, teams need to evaluate who is impacted by their project and what this influence means to the person, group or society as a whole.

In case you want to do research on human subjects (this includes for example interviews or surveys) we have included a questionnaire to identify possible issues that you need to evaluate if your experimental questions validates the ethical concern of pursuing this experiment. This questionnaire should give you an idea about the broad range of considerations necessary when you want to conduct research on human subjects. We do still encourage you to always get approval by your institute's ethic council if you would like to conduct such experiments.

If you would like to learn more about bioethics, we recommend to take a look at the NIH or UNESCO bioethic websites (NIH: <https://www.niehs.nih.gov/research/resources/bioethics/index.cfm> UNESCO: <https://en.unesco.org/themes/ethics-science-and-technology/bioethics>).

23. List all people/groups that are potentially influenced by your research or outcomes of your research and describe the impact of your project on these people/groups. *

Our project will contribute to large-fragment DNA manipulation. This will help a broader range of biotechnology practitioners, enabling them to modify DNA more easily. In the long term, it may also benefit patients with genetic diseases in the decades to come.

24. Do you see any ethical issues arising from your project? If so, indicate why. *

No, we just develop new tools.

25. Do you plan to conduct research on humans or human samples (this includes questionnaires, interviews, public engagement)? *

No

26. If yes, does your experiment fulfil all requirements to be categorized as research with low potential to cause ethical issues? (A good guideline is the questionnaire below. If you can answer all questions on the questionnaire below with no, your research probably has a low potential to be the cause of ethical issues.) *

This questionnaire is derived from the basic questionnaire of the ethics council of Bielefeld University.

Please answer all 12 questions by ticking yes or no as appropriate:	DGPs*	yes	no
1. Will members of a vulnerable group or people who cannot give their own consent participate in this study (e.g., children and adolescents under 16 years of age, people with a learning disability, people in psychotherapeutic treatment)?	3 (b)		
2. Will it be necessary that people participate in this study without having been informed about this previously or without having given their consent to participate (e.g., as in covert observation)?	6		
3. Will the study involve covert observation or any other method that precludes informed consent, full debriefing, or the opportunity for participants to have their data deleted?	3, 9		
4. Will the study feature questions about topics that are of an intimate nature or that participants may perceive as stigmatizing (e.g., questions pertaining to illegal or deviant behavior or to sexual preferences)?	3 (d)		
5. Does the study include an active deception of participants or will information be deliberately withheld from participants? (This does not apply to withholding the study hypothesis.)	8		
6. Is there a risk that the study may cause psychological stress, fear, exhaustion, or other negative effects in participants to an extent that they would not normally encounter in their daily life?	3 (d), 9		
7. Is there a risk that the study may cause participants to experience pain or more than mild discomfort?	3 (d), 9		
8. Will the participants be given drugs, placebos, or other substances to ingest (e.g., food, beverages, vitamins), or will the participants be subjected to any invasive or potentially harmful procedures?	3 (d), 3 (e), 8, 9		
9. Will video or audio recordings be taken without prior consent by the participants ?	3, 4		
10. Will any bodily substances of participants be sampled (blood, saliva, etc.)?	3, 4		
11. Will participants receive a payment of more than 10 Euros <i>per hour</i> for their participation?	7		
12. Is there a conflict of interest for the applicant or applicants because of (a) economic or personal connections to a contracting entity or a collaborator whose interests may be affected positively or negatively by results of the research, or (b) any other factor(s) that might affect the applicants' independent scientific judgment?	–		

Note: * This column points out particularly relevant subsections of section 7.3 of the document "Berufsethische Richtlinien der DGPs und des BDP". See website of the EUB.

N.A.

Thank
you!

We hope this responsible research form gave you some idea how complex safety, security and ethical considerations for a research project are and that you learned a little bit more about your project by filling out this form. Two last questions:

We would like to ask you about the time your team needed to fill out the form and if you would consider the effort as too much.

Would it be okay if we use your answers to learn more about risk assessment of directed evolution projects? We might give a talk at the festival where we would like to talk about case studies or publish about the iDEC risk assessment resources. This does not include your email address or PI information and we would of course anonymize all answers, but someone might be able to connect projects to answers given in the form.

27. How much time did you and your team need to consider risks and to fill out the form. Would you consider this time as too long, too short or adequate?

Adequate. It took me about 4 hours.

28. Do you give consent for your answers in this form to be anonymously used for talks and publications by the iDEC Commission for Responsible Research? *

Mark only one oval.

☒ Yes

☐ No